

An Infinite Counterfamily to a Set Identity in a Recursive Collatz Sieve

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Abstract

The proof of Lemma 3.1 in Ansari's recursive Collatz sieve uses a displayed set identity to complete an induction. At its $n = 2$ specialization, every integer $75 + 108t$, with t nonnegative, belongs to the proposed set difference but not to the next sieve. This gives an infinite counterfamily to that identity. The arithmetic statement has also been checked in Lean. The result identifies a failure of the displayed induction step; it does not decide whether the paper's recursive sufficiency conclusion admits another proof.

The Set Identity

In the proof of Lemma 3.1 of Ansari [1], the final equality in the induction on printed page 478 identifies the next sieve with a larger digit set after subtracting an exceptional set. The $n = 2$ case is enough to test this equality without any convention about an empty string of digits. Throughout, k and t range over the nonnegative integers.

Write $E(k, a_2, a_1, a_0)$ for the common digit expression. The definitions on pages 477–478 of [1], specialized to $n = 2$, are:

$$E(k, a_2, a_1, a_0) = 108k + 36a_2 + 12a_1 + 4a_0 + 3$$

$$F_3 = \{E(k, a_2, a_1, a_0) : a_0, a_1, a_2 \in \{0, 1\}\}$$

$$F'_2 = \{E(k, a_2, a_1, a_0) : a_0 \in \{0, 1\}, a_1, a_2 \in \{0, 1, 2\}\}$$

$$A'_2 = \{108k + 99 + 4a_0 : a_0 \in \{0, 1\}\}$$

The equality asserted at the end of that induction becomes

$$F_3 = F'_2 \setminus A'_2.$$

An Infinite Counterfamily

Proposition. For every $t \geq 0$, the integer $x_t = 75 + 108t$ belongs to $F'_2 \setminus A'_2$ and does not belong to F_3 . Consequently, the displayed equality is false.

Proof. Set $k = t$, $a_2 = 2$, and $a_1 = a_0 = 0$. Then $E(t, 2, 0, 0) = 75 + 108t$, so x_t belongs to F'_2 . Modulo 108, the only residues of A'_2 are 99 and 103. The residues of F_3 are 3, 7, 15, 19, 39, 43, 51, and 55. These lists are complete: each bounded digit offset lies between 0 and 107. Since x_t has residue 75, it belongs to neither A'_2 nor F_3 . This holds for every nonnegative t . ■

The same family also lies in the preceding sieve F_2 : $75 + 108t = 36(2 + 3t) + 3$, with $a_1 = a_0 = 0$ in the definition of F_2 .

The Digit Set Difference

The difference between the two sides is visible in the digit restrictions. Within F'_2 , let B_2 contain the elements represented with $a_2 = 2$ and let C_2 contain those represented with $a_1 = 2$. The bounded representation is unique: $(x - 3)/4 = 27k + a_0 + 3a_1 + 9a_2$, where $a_0 \in \{0, 1\}$ and $a_1, a_2 \in \{0, 1, 2\}$. The digit sum is less than 27. Thus the elements admitted to F'_2 but absent from F_3 have at least one of those digits equal to 2, while A'_2 selects only the case in which both equal 2. Equivalently,

$$F_3 = F'_2 \setminus (B_2 \cup C_2), A'_2 = B_2 \cap C_2.$$

The counterfamily has $a_2 = 2$ and $a_1 = 0$, so subtracting A'_2 leaves it in the right-hand side of the printed identity.

Formal Verification and Scope

The Lean 4 companion `AnsariSieveIdentity.lean` defines the four specialized predicates over natural numbers. It proves the counterfamily's membership in F_2 and F'_2 and nonmembership in A'_2 and F_3 , combines these claims in one theorem, and proves the parameterization injective and unbounded. It also proves the negation of the universal membership equivalence:

$$\neg (\forall x : \mathbb{N}, F_3(x) \leftrightarrow (F'_2(x) \wedge \neg A'_2(x))).$$

All eight theorem declarations compile from a clean package under Lean 4.33.1. Their axiom reports list only `propext` and `Quot.sound`. This verifies the arithmetic consequence of the encoded definitions. Matching those definitions to the printed article was a separate manual check; the article itself was not formalized.

The counterfamily invalidates the displayed equality and therefore that particular last step of the induction. It does not show that the Collatz conjecture is false, nor does it rule out another argument for the paper's recursive sufficiency claim. Related public audits discuss defects at nearby sieve levels [2, 3].

References

- [1] Mohammad Ansari, "Recursive sufficiency for the Collatz conjecture and computational verification," *Notes on Number Theory and Discrete Mathematics* 31, no. 3 (2025): 471–480. DOI: 10.7546/nntdm.2025.31.3.471-480.
- [2] jkolantree, "Collatz Recursive Sufficiency Audit," §2, BSC repository, revision 2944e11ca39c39bf95533342c765379f6bb7da02.
- [3] Sodelin, "Sufficiency Rank Audit," §3, Collatz-Conjecture-Work repository, revision 33922a42e86646258d227d1e19c6cf7546a2f548.